

4. Consider The Sun

(a) What is the Sun's escape velocity?

(b) Imagine now it condensed such that it has the same mass, but its radius is now 4 km. What is The Sun's new escape velocity?

(c) What does this imply about what kind of celestial body the sun would now be?

5. Derive the equation for the amount of work required for an object of mass m to escape a celestial body of mass M and radius r

(a) Does this result make sense? Explain why or why not: